```
┌──(venv)─(root㉿kali)-[/home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/scripts]
└─# ./Run_Pipeline.sh

  ╔══════════════════════════════════════════════════════╗
  ║   DNS Traffic Anomaly Detection — Pipeline Runner    ║
  ╚══════════════════════════════════════════════════════╝

  ℹ  Script dir : /home/kali/Downloads/DNS-_raffic-_Anomaly_Detection/scripts
  ℹ  Data dir   : /home/kali/Downloads/DNS-_raffic-_Anomaly_Detection/data
  ℹ  Model mem  : /home/kali/Downloads/DNS-_raffic-_Anomaly_Detection/model_memory
  ℹ  Log file   : /home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/logs/pipeline_20260314_061448.log

══ Checking trained model memory
  ✅  Path A models ready (trained: 2026-03-13T14:14:26Z)
  ✅  Path B model ready (trained: 2026-03-13T14:14:26Z)
  ℹ  Path A models loaded from memory
  ℹ  Path B model loaded from memory

  📋  Model memory info:
     Last trained    : 2026-03-13T14:14:26Z
     Trained on      : 11430 samples (5451 attacks + 5979 benign)
     Training runs   : 2 total

══ Preflight checks
  ✅  python3: Python 3.13.3
  ✅  All script files present
  ✅  Found 2 DNS log file(s)
  ✅  Python packages ready

══ STEP 1 — Week 3: Feature Extraction (new traffic)
  ℹ  Parsing new DNS logs from: /home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/technitium-dns/data/dns_logs
================================================================================
DNS FEATURE EXTRACTION FROM TECHNITIUM LOGS - WEEK 3
Enhanced for Attack Tool Detection
================================================================================

📁 Log directory: /home/kali/Downloads/DNS-_raffic-_Anomaly_Detection/technitium-
dns/data/dns_logs
📁 Output directory: /home/kali/Downloads/DNS-_raffic-_Anomaly_Detection/data
```

```
Found 2 log files to process
Processing log file: /home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/technitium-dns/data/dns_logs/2026-03-13.log
  Completed: 175939 lines processed, 37242 DNS queries extracted
Processing log file: /home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/technitium-dns/data/dns_logs/2026-03-14.log
  Completed: 46 lines processed, 0 DNS queries extracted

================================================================================
DNS FEATURE EXTRACTION SUMMARY - WEEK 3 (ATTACK DETECTION FOCUS)
================================================================================
Total queries processed: 37242

🔴 ATTACK DETECTION SUMMARY:
   Suspicious queries: 11616 (31.2%)

   Attack Type Breakdown:
     • ENCODING: 8543
     • AMPLIFICATION: 3707
     • HIGH_ENTROPY: 225
     • NXDOMAIN_FLOOD: 32

📅 Date range: 2026-03-13 to 2026-03-13
   Unique domains: 37242
   Unique client IPs: 1

🎯 ATTACK TOOL PATTERNS DETECTED:
   NXDOMAIN flood patterns: 32
   Random subdomain patterns: 0
   Amplification attempts: 3707
   Cache poisoning attempts: 0
   High entropy domains: 225
   Encoding patterns: 13764

📊 FEATURE STATISTICS (for baseline establishment):
------------------------------------------------------------


entropy:
  Mean: 3.4073
  Std: 0.3071
  Min: 2.2842
  Max: 4.5094
```

```
subdomain_depth:
  Mean: 2.5137
  Std: 0.7264
  Min: 2.0000
  Max: 7.0000

query_length:
  Mean: 22.1198
  Std: 6.6658
  Min: 11.0000
  Max: 71.0000

numeric_ratio:
  Mean: 0.0164
  Std: 0.0401
  Min: 0.0000
  Max: 0.4783

first_label_entropy:
  Mean: 2.3617
  Std: 0.9582
  Min: 0.0000
  Max: 4.4410

first_label_length:
  Mean: 8.4696
  Std: 5.4441
  Min: 1.0000
  Max: 53.0000

⚠ TOP 10 MOST SUSPICIOUS DOMAINS:
  1. production-staging-merchant.extnp.lab.local
     Attacks: AMPLIFICATION,HIGH_ENTROPY,ENCODING | Entropy: 4.15 | Length: 43
  2. powerbossequipment-stage-www.lab.local
     Attacks: AMPLIFICATION,HIGH_ENTROPY,ENCODING | Entropy: 4.06 | Length: 38
  3. keepstock-vending-device-bff.lab.local
     Attacks: AMPLIFICATION,HIGH_ENTROPY,ENCODING | Entropy: 4.04 | Length: 38
  4. qahigthirdpartyassessmentprogram.lab.local
     Attacks: NXDOMAIN_FLOOD,HIGH_ENTROPY,ENCODING | Entropy: 4.03 | Length: 42
  5. dtprod2-billing-invoice.dtprod2.dsl.aws.lab.local
     Attacks: AMPLIFICATION,HIGH_ENTROPY,ENCODING | Entropy: 4.03 | Length: 49
  6. ecostruxure-power-build-okkenblokset.lab.local
     Attacks: AMPLIFICATION,HIGH_ENTROPY,ENCODING | Entropy: 4.03 | Length: 46
  7. filermgmt-enterprise.appsdmz.edgar.lab.local
```

```
     Attacks: AMPLIFICATION,HIGH_ENTROPY,ENCODING | Entropy: 4.02 | Length: 44
  8. stg-stage-productionservices.lab.local
     Attacks: NXDOMAIN_FLOOD,HIGH_ENTROPY,ENCODING | Entropy: 4.02 | Length: 38
  9. honda-prod-test-01-fwuzdnft.lab.local
     Attacks: AMPLIFICATION,HIGH_ENTROPY | Entropy: 4.19 | Length: 37
  10. dtprod2-myaccount-knowledgebase.dtprod2.dsl.aws.lab.local
     Attacks: HIGH_ENTROPY,ENCODING | Entropy: 4.17 | Length: 57

✅ Features exported to JSON: /home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/data/week3_features_all.json
   Total records: 37242

✅ Attack summary exported to: /home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/data/week3_attack_summary.json

✅ Features exported to JSON: /home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/data/week3_suspicious_domains.json
   Total records: 11616

⚠  Found 11616 suspicious domains for Week 4 testing

✅ Project summary saved to: /home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/data/week3_project_summary.json

📂 Files in output directory:
  • week3_attack_summary.json (707 bytes)
  • week3_features_all.json (38,679,812 bytes)
  • week3_features_labeled.json (12,548,895 bytes)
  • week3_features_with_ml.json (57,433,607 bytes)
  • week3_project_summary.json (318 bytes)
  • week3_suspicious_domains.json (12,173,293 bytes)
  ✅ Week 3 complete — 37242 records extracted

══ STEP 2 — ML Bridge: Applying trained model memory to new traffic
  ℹ Using models trained on: 2026-03-13T14:14:26Z
  ℹ Path B: active (Trafficformer reconstruction anomaly detection)
WARNING: All log messages before absl::InitializeLog() is called are written to
STDERR
I0000 00:00:1773504895.465128   12586 port.cc:153] oneDNN custom operations are
on. You may see slightly different numerical results due to floating-point round-
off errors from different computation orders. To turn them off, set the
environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
I0000 00:00:1773504895.465308   12586 cudart_stub.cc:31] Could not find cuda
drivers on your machine, GPU will not be used.
```

```
I0000 00:00:1773504895.506007   12586 cpu_feature_guard.cc:227] This TensorFlow
binary is optimized to use available CPU instructions in performance-critical
operations.
To enable the following instructions: AVX2 AVX_VNNI FMA, in other operations,
rebuild TensorFlow with the appropriate compiler flags.
WARNING: All log messages before absl::InitializeLog() is called are written to
STDERR
I0000 00:00:1773504896.295101   12586 port.cc:153] oneDNN custom operations are
on. You may see slightly different numerical results due to floating-point round-
off errors from different computation orders. To turn them off, set the
environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
I0000 00:00:1773504896.295317   12586 cudart_stub.cc:31] Could not find cuda
drivers on your machine, GPU will not be used.
E0000 00:00:1773504897.147044   12586 cuda_platform.cc:52] failed call to cuInit:
INTERNAL: CUDA error: Failed call to cuInit: UNKNOWN ERROR (303)

======================================================================
ML BRIDGE — Connecting Week 3 → ML Models → Week 4
======================================================================
Input  : /home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/data/week3_features_all.json
Output : /home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/data/week3_features_with_ml.json

[*] Loaded 37242 Week 3 records

── Path A: RF + Transformer (8-feature ensemble) ─────────────
   Using ensemble threshold: 0.4316
[*] Reading Week 3 features JSON: /home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/data/week3_features_all.json
[*] Loaded 37242 Week 3 feature records.
   Path A results: 7590 attack  |  29652 benign

── Path B: Trafficformer (sequence reconstruction) ───────────

⚠ Path B skipped: Model not found: model/dns_trafficformer.keras

── Merging ML scores ──────────────────────────────────────────

── Bridge Summary ─────────────────────────────────────────────
   Total records      : 37242
   ML attacks flagged : 7590 (20.4%)
   HIGH confidence    : 0
   MEDIUM confidence  : 0
```

```
   LOW confidence     : 2847
   CLEAN              : 34395

   New columns added to Week 3 features:
     + ml_score_A
     + ml_label_A
     + ml_verdict_A
     + ml_score_B
     + ml_label_B
     + ml_combined_score
     + ml_combined_label
     + ml_combined_verdict
     + ml_confidence
     + ml_threat_score

✅ Bridge output saved → /home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/data/week3_features_with_ml.json
   Next step: python Week4_detection_engine.py --input /home/kali/Downloads/DNS-
_raffic-_Anomaly_Detection/data/week3_features_with_ml.json
  ✅ ML bridge complete — new traffic scored against trained models

══ STEP 3 — Week 4: Detection Engine
  ℹ  Running 6 detection rules (4 Sentinel + Z-score + 2 ML rules)

================================================================================
WEEK 4 (UPDATED) — UNIFIED DNS DETECTION ENGINE
================================================================================
Rule: Excessive NXDOMAIN DNS Queries  [Microsoft Sentinel ASIM DNS Solution]
Rule: Multiple Errors for Same DNS Query  [Microsoft Sentinel ASIM DNS Solution]
Rule: DNS Tunneling Detection  [Project Requirement]
Rule: Deep Subdomain Detection  [Project Requirement]

ML Rules (active if ml_bridge.py output is used):
  Rule 5: ML Model Attack Detection
  Rule 6: ML High Threat Score  (threshold: 0.7)

Z-score threshold: 2.0 σ  [NetSleuthXplorer]
Microsoft Defender TI: Disabled
================================================================================

📂 Loading features from: /home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/data/week3_features_with_ml.json
✅ ML scores detected — ML rules (5 & 6) will be active
✅ Loaded 37242 DNS queries
```

```
   Date range: 2026-03-13 23:45:18 to 2026-03-13 23:45:26

📊 Building statistical baseline for Z-score detection...
✅ Built baseline for 1 unique IPs

================================================================================
DETECTION ENGINE EXECUTION
================================================================================

🔍 [Rule 1] Running: Excessive NXDOMAIN DNS Queries
   📊 Found 547 NXDOMAIN responses
   🚨 Alerts generated: 1

🔍 [Rule 2] Running: Multiple Errors for Same DNS Query
   📊 Found 547 error responses
   🚨 Alerts generated: 0

🔍 [Rule 3] Running: DNS Tunneling Detection
   📊 Found 603 potential tunneling queries
   🚨 Alerts generated: 100

🔍 [Rule 4] Running: Deep Subdomain Detection
   📊 Found 112 queries with deep subdomains
   🚨 Alerts generated: 50

🔍 [Z-score] Running: Statistical Anomaly Detection
   🚨 Statistical alerts: 6437

🔍 [Rule 5] Running: ML Model Attack Detection
   📊 Found 7590 ML-flagged queries
   🚨 Alerts generated: 200

🔍 [Rule 6] Running: ML High Threat Score
   📊 Found 0 queries with threat score >= 0.7

🔍 Enriching 1 IPs with Microsoft Defender TI...
Enriching IPs: 100%|██████████| 1/1 [00:00<00:00, 35848.75IP/s]
⚠ MDTI credentials not configured. Skipping API calls.
⚠ MDTI credentials not configured. Skipping API calls.

================================================================================
DETECTION SUMMARY
================================================================================
Total queries analyzed : 37242
```

```
Total alerts generated : 6788
ML Rules (5 & 6)       : ✅ Active

Alerts by Rule:
  - Excessive NXDOMAIN DNS Queries: 1
  - DNS Tunneling Detection: 100
  - Deep Subdomain Detection: 50
  - ML Model Attack Detection: 200

Alerts by Severity:
  - High: 101
  - Medium: 250

Top Offending IPs:
  - 192.168.44.144: 301

Top Suspicious Domains:
  - cafeskthumb-phinf.lab.local: 2
  - sadmin-internal-gateway-api.dev.lab.local: 2
  - sadmin-external-gateway-api.dev.lab.local: 2
  - identity-2.uk-south.iam.test.cloud.lab.local: 2
  - N/A: 1
================================================================================

✅ Alerts exported to: /home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/data/week4_unified_alerts.json
✅ Alerts CSV exported to: /home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/data/week4_unified_alerts.csv

✅ Detection complete.
   Rule 1–4: Microsoft Sentinel rules
   Z-score:  NetSleuthXplorer statistical detection
   Rule 5–6: ML detection (Path A: RF+Transformer, Path B: Trafficformer)
  ✅ Week 4 complete — 6788 total alerts (1226 High, 5562 Medium)

══════════════════════════════════════════════════════

  ✅  DETECTION COMPLETE  (0m 17s)
══════════════════════════════════════════════════════


Results:
  🚨 Total alerts      : 6788
  🔴 High severity     : 1226
  🟠 Medium severity   : 5562
```

```
Output files:
  📄 Week 3 features    : /home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/data/week3_features_all.json
  🤖 ML-enriched data   : /home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/data/week3_features_with_ml.json
  🚨 Alerts (JSON)      : /home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/data/week4_unified_alerts.json
  📊 Alerts (CSV)       : /home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/data/week4_unified_alerts.csv
  📋 Pipeline log      : /home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/logs/pipeline_20260314_061448.log

  Detection breakdown by rule:
    🤖 ML Model Attack Detection: 200
    🔍 DNS Tunneling Detection: 100
    🔍 Deep Subdomain Detection: 50
    🔍 Excessive NXDOMAIN DNS Queries: 1

Model memory used:
  Stored in: /home/kali/Downloads/DNS-_raffic-_Anomaly_Detection/model_memory
  Trained on : 2026-03-13T14:14:26Z
  Samples    : 11430 (5451 attack + 5979 benign)

  To retrain on new data:
  ./Train_Models.sh --retrain


┌──(venv)─(root㉿kali)-[/home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/scripts]
└─# ./Train_Models.sh --retrain



  ╔═══════════════════════════════════════════════════════════════╗
  ║   DNS Anomaly Detection — ML Model Trainer v2                 ║
  ║   Sources: CSV · JSON · PCAP · LOG                            ║
  ║   Memory:  full provenance saved for pipeline awareness        ║
  ╚═══════════════════════════════════════════════════════════════╝

  ℹ  Script dir     : /home/kali/Downloads/DNS-_raffic-_Anomaly_Detection/scripts
  ℹ  Model memory   : /home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/model_memory
  ℹ  Data dir       : /home/kali/Downloads/DNS-_raffic-_Anomaly_Detection/data
  ℹ  Log file       : /home/kali/Downloads/DNS-_raffic-
```

```
_Anomaly_Detection/logs/training_20260314_061518.log

══ Checking existing trained models
  ✅ Path A models exist (trained: 2026-03-13T14:14:26Z)
  ✅ Path B model exists (trained: 2026-03-13T14:14:26Z)
  ℹ  Training needed — Path A: true  |  Path B: true

══ Preflight checks
  ✅ python3: Python 3.13.3
  ✅ tshark: Running as user "root" and group "root". This could be dangerous.
  ✅ All required scripts present
  ✅ Python packages ready

══ SOURCE DISCOVERY — scanning for training data
  ⚠  No files specified — falling back to default DNS log directory
  ℹ  Scanning: /home/kali/Downloads/DNS-_raffic-_Anomaly_Detection/technitium-
dns/data/dns_logs
     ↳ LOG  : /home/kali/Downloads/DNS-_raffic-_Anomaly_Detection/technitium-
dns/data/dns_logs/2026-03-13.log
     ↳ LOG  : /home/kali/Downloads/DNS-_raffic-_Anomaly_Detection/technitium-
dns/data/dns_logs/2026-03-14.log

  ✅ Source files discovered:
  ✅   CSV  : 0 file(s)
  ✅   JSON : 0 file(s)
  ✅   PCAP : 0 file(s)
  ✅   LOG  : 2 file(s)

══ STEP 0-A — Pre-processing raw LOG files
  ℹ  Found 2 raw log file(s) — extracting features via DNS_feature_extractor.py
     ↳ Extracting: /home/kali/Downloads/DNS-_raffic-_Anomaly_Detection/technitium-
dns/data/dns_logs/2026-03-13.log → /home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/data/extracted_2026-03-13_1773504921.json
================================================================================
DNS FEATURE EXTRACTION FROM TECHNITIUM LOGS - WEEK 3
Enhanced for Attack Tool Detection
================================================================================

📁 Single file mode: /home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/technitium-dns/data/dns_logs/2026-03-13.log
Processing log file: /home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/technitium-dns/data/dns_logs/2026-03-13.log
  Completed: 175939 lines processed, 37242 DNS queries extracted
```

```
================================================================================
DNS FEATURE EXTRACTION SUMMARY - WEEK 3 (ATTACK DETECTION FOCUS)
================================================================================
Total queries processed: 37242

🔴 ATTACK DETECTION SUMMARY:
   Suspicious queries: 11616 (31.2%)

   Attack Type Breakdown:
     • ENCODING: 8543
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     • HIGH_ENTROPY: 225
     • NXDOMAIN_FLOOD: 32

📅 Date range: 2026-03-13 to 2026-03-13
   Unique domains: 37242
   Unique client IPs: 1

🎯 ATTACK TOOL PATTERNS DETECTED:
   NXDOMAIN flood patterns: 32
   Random subdomain patterns: 0
   Amplification attempts: 3707
   Cache poisoning attempts: 0
   High entropy domains: 225
   Encoding patterns: 13764

📊 FEATURE STATISTICS (for baseline establishment):
------------------------------------------------------------


entropy:
  Mean: 3.4073
  Std: 0.3071
  Min: 2.2842
  Max: 4.5094

subdomain_depth:
  Mean: 2.5137
  Std: 0.7264
  Min: 2.0000
  Max: 7.0000

query_length:
  Mean: 22.1198
  Std: 6.6658
```

```
  Min: 11.0000
  Max: 71.0000

numeric_ratio:
  Mean: 0.0164
  Std: 0.0401
  Min: 0.0000
  Max: 0.4783

first_label_entropy:
  Mean: 2.3617
  Std: 0.9582
  Min: 0.0000
  Max: 4.4410

first_label_length:
  Mean: 8.4696
  Std: 5.4441
  Min: 1.0000
  Max: 53.0000

⚠ TOP 10 MOST SUSPICIOUS DOMAINS:
  1. production-staging-merchant.extnp.lab.local
     Attacks: AMPLIFICATION,HIGH_ENTROPY,ENCODING | Entropy: 4.15 | Length: 43
  2. powerbossequipment-stage-www.lab.local
     Attacks: AMPLIFICATION,HIGH_ENTROPY,ENCODING | Entropy: 4.06 | Length: 38
  3. keepstock-vending-device-bff.lab.local
     Attacks: AMPLIFICATION,HIGH_ENTROPY,ENCODING | Entropy: 4.04 | Length: 38
  4. qahigthirdpartyassessmentprogram.lab.local
     Attacks: NXDOMAIN_FLOOD,HIGH_ENTROPY,ENCODING | Entropy: 4.03 | Length: 42
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     Attacks: AMPLIFICATION,HIGH_ENTROPY,ENCODING | Entropy: 4.03 | Length: 49
  6. ecostruxure-power-build-okkenblokset.lab.local
     Attacks: AMPLIFICATION,HIGH_ENTROPY,ENCODING | Entropy: 4.03 | Length: 46
  7. filermgmt-enterprise.appsdmz.edgar.lab.local
     Attacks: AMPLIFICATION,HIGH_ENTROPY,ENCODING | Entropy: 4.02 | Length: 44
  8. stg-stage-productionservices.lab.local
     Attacks: NXDOMAIN_FLOOD,HIGH_ENTROPY,ENCODING | Entropy: 4.02 | Length: 38
  9. honda-prod-test-01-fwuzdnft.lab.local
     Attacks: AMPLIFICATION,HIGH_ENTROPY | Entropy: 4.19 | Length: 37
  10. dtprod2-myaccount-knowledgebase.dtprod2.dsl.aws.lab.local
     Attacks: HIGH_ENTROPY,ENCODING | Entropy: 4.17 | Length: 57

✅ Features exported to JSON: /home/kali/Downloads/DNS-_raffic-
```

```
_Anomaly_Detection/data/extracted_2026-03-13_1773504921.json
   Total records: 37242
     ↳ Extracting: /home/kali/Downloads/DNS-_raffic-_Anomaly_Detection/technitium-dns/data/dns_logs/2026-03-14.log → /home/kali/Downloads/DNS-_raffic-_Anomaly_Detection/data/extracted_2026-03-14_1773504924.json
================================================================================
DNS FEATURE EXTRACTION FROM TECHNITIUM LOGS - WEEK 3
Enhanced for Attack Tool Detection
================================================================================

📁 Single file mode: /home/kali/Downloads/DNS-_raffic-_Anomaly_Detection/technitium-dns/data/dns_logs/2026-03-14.log
Processing log file: /home/kali/Downloads/DNS-_raffic-_Anomaly_Detection/technitium-dns/data/dns_logs/2026-03-14.log
  Completed: 46 lines processed, 0 DNS queries extracted

❌ No features extracted.
  ✅ Log pre-processing done — 1 JSON file(s) produced

══ STEP 0-B — Pre-processing PCAP files
  ℹ  No PCAP files — skipping PCAP pre-processing

══ STEP 0-C — Merging all sources into one feature dataset
  [json]  37242 records  ← /home/kali/Downloads/DNS-_raffic-_Anomaly_Detection/data/extracted_2026-03-13_1773504921.json

  Merged 37242 total records from 1 source file(s)
  Provenance saved → /home/kali/Downloads/DNS-_raffic-_Anomaly_Detection/model_memory/training_provenance.json
  ✅ Merged dataset: 37242 records → /home/kali/Downloads/DNS-_raffic-_Anomaly_Detection/data/merged_features_all.json

══ STEP 1 — Generate labels from merged dataset
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1773504925.657837   12980 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
I0000 00:00:1773504925.658098   12980 cudart_stub.cc:31] Could not find cuda drivers on your machine, GPU will not be used.
I0000 00:00:1773504925.696112   12980 cpu_feature_guard.cc:227] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations.
```

```
To enable the following instructions: AVX2 AVX_VNNI FMA, in other operations,
rebuild TensorFlow with the appropriate compiler flags.
WARNING: All log messages before absl::InitializeLog() is called are written to
STDERR
I0000 00:00:1773504926.432332   12980 port.cc:153] oneDNN custom operations are
on. You may see slightly different numerical results due to floating-point round-
off errors from different computation orders. To turn them off, set the
environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
I0000 00:00:1773504926.432564   12980 cudart_stub.cc:31] Could not find cuda
drivers on your machine, GPU will not be used.

======================================================================
ML BRIDGE — Connecting Week 3 → ML Models → Week 4
======================================================================
Input  : /home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/data/merged_features_all.json
Output : /home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/data/merged_features_labeled.json

[*] Loaded 37242 Week 3 records

[*] LABEL ONLY MODE — generating labels from attack_count
   Labels generated: 25626 benign  |  11616 attack
   Attack rate: 31.2%

✅ Labeled data saved → /home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/data/merged_features_labeled.json
   Use for training: python Model.py --train /home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/data/merged_features_labeled.json
  ✅ Labels: attacks=11616  benign=25626  total=37242
  ✅ Attack types found: unknown

══ STEP 2 — Training Path A: RF + Transformer (8 features)
  ℹ  Learns WHICH patterns are attacks vs benign (supervised)
  ℹ  Dataset : /home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/data/merged_features_labeled.json (37242 records)
WARNING: All log messages before absl::InitializeLog() is called are written to
STDERR
I0000 00:00:1773504930.061859   13029 port.cc:153] oneDNN custom operations are
on. You may see slightly different numerical results due to floating-point round-
off errors from different computation orders. To turn them off, set the
environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
I0000 00:00:1773504930.062307   13029 cudart_stub.cc:31] Could not find cuda
drivers on your machine, GPU will not be used.
```

```
I0000 00:00:1773504930.105021   13029 cpu_feature_guard.cc:227] This TensorFlow
binary is optimized to use available CPU instructions in performance-critical
operations.
To enable the following instructions: AVX2 AVX_VNNI FMA, in other operations,
rebuild TensorFlow with the appropriate compiler flags.
WARNING: All log messages before absl::InitializeLog() is called are written to
STDERR
I0000 00:00:1773504931.170649   13029 port.cc:153] oneDNN custom operations are
on. You may see slightly different numerical results due to floating-point round-
off errors from different computation orders. To turn them off, set the
environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
I0000 00:00:1773504931.170918   13029 cudart_stub.cc:31] Could not find cuda
drivers on your machine, GPU will not be used.
E0000 00:00:1773504932.705180   13029 cuda_platform.cc:52] failed call to cuInit:
INTERNAL: CUDA error: Failed call to cuInit: UNKNOWN ERROR (303)

[*] Loading training data: /home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/data/merged_features_labeled.json
[*] Reading Week 3 features JSON: /home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/data/merged_features_labeled.json
[*] Loaded 37242 Week 3 feature records.
[*] Dataset: 37242 records  |  Benign: 25626  |  Attack: 11616
Saved → dns_scaler.pkl

── Random Forest ──────────────────────────────────────────────
              precision    recall  f1-score   support

      benign       0.98      0.99      0.98      7688
      attack       0.97      0.95      0.96      3485

    accuracy                           0.97     11173
   macro avg       0.97      0.97      0.97     11173
weighted avg       0.97      0.97      0.97     11173

Saved → dns_tunnel_rf.pkl

── Transformer (Binary Classifier) ───────────────────────────
Model: "functional"
┏━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━━━┓
┃ Layer (type)        ┃ Output Shape      ┃    Param # ┃ Connected to      ┃
┡━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━━━┩
│ input_layer         │ (None, 10, 8)     │          0 │ -                 │
│ (InputLayer)        │                   │            │                   │
├─────────────────────┼───────────────────┼────────────┼───────────────────┤
```

layer_normalization (LayerNormalizatio…	(None, 10, 8)	16	input_layer[0][0]
multi_head_attenti… (MultiHeadAttentio…	(None, 10, 8)	2,248	layer_normalizat… layer_normalizat…
dropout_1 (Dropout)	(None, 10, 8)	0	multi_head_atten…
add (Add)	(None, 10, 8)	0	dropout_1[0][0], input_layer[0][0]
layer_normalizatio… (LayerNormalizatio…	(None, 10, 8)	16	add[0][0]
conv1d (Conv1D)	(None, 10, 64)	576	layer_normalizat…
dropout_2 (Dropout)	(None, 10, 64)	0	conv1d[0][0]
conv1d_1 (Conv1D)	(None, 10, 8)	520	dropout_2[0][0]
add_1 (Add)	(None, 10, 8)	0	conv1d_1[0][0], add[0][0]
layer_normalizatio… (LayerNormalizatio…	(None, 10, 8)	16	add_1[0][0]
multi_head_attenti… (MultiHeadAttentio…	(None, 10, 8)	2,248	layer_normalizat… layer_normalizat…
dropout_4 (Dropout)	(None, 10, 8)	0	multi_head_atten…
add_2 (Add)	(None, 10, 8)	0	dropout_4[0][0], add_1[0][0]
layer_normalizatio… (LayerNormalizatio…	(None, 10, 8)	16	add_2[0][0]
conv1d_2 (Conv1D)	(None, 10, 64)	576	layer_normalizat…
dropout_5 (Dropout)	(None, 10, 64)	0	conv1d_2[0][0]
conv1d_3 (Conv1D)	(None, 10, 8)	520	dropout_5[0][0]
add_3 (Add)	(None, 10, 8)	0	conv1d_3[0][0],

```
│                     │                   │            │ add_2[0][0]       │
├─────────────────────┼───────────────────┼────────────┼───────────────────┤
│ global_average_poo… │ (None, 8)         │          0 │ add_3[0][0]       │
│ (GlobalAveragePool… │                   │            │                   │
├─────────────────────┼───────────────────┼────────────┼───────────────────┤
│ dense (Dense)       │ (None, 64)        │        576 │ global_average_p… │
├─────────────────────┼───────────────────┼────────────┼───────────────────┤
│ dropout_6 (Dropout) │ (None, 64)        │          0 │ dense[0][0]       │
├─────────────────────┼───────────────────┼────────────┼───────────────────┤
│ dense_1 (Dense)     │ (None, 1)         │         65 │ dropout_6[0][0]   │
└─────────────────────┴───────────────────┴────────────┴───────────────────┘
 Total params: 7,393 (28.88 KB)
 Trainable params: 7,393 (28.88 KB)
 Non-trainable params: 0 (0.00 B)
Epoch 1/30
815/815 ━━━━━━━━━━━━━━━━━━━━ 9s 8ms/step - accuracy: 0.7257 - auc: 0.6415 - loss: 0.5787 - val_accuracy: 0.7293 - val_auc: 0.6476 - val_loss: 0.5720
Epoch 2/30
815/815 ━━━━━━━━━━━━━━━━━━━━ 6s 8ms/step - accuracy: 0.7296 - auc: 0.6508 - loss: 0.5732 - val_accuracy: 0.7298 - val_auc: 0.6484 - val_loss: 0.5731
Epoch 3/30
815/815 ━━━━━━━━━━━━━━━━━━━━ 6s 8ms/step - accuracy: 0.7293 - auc: 0.6546 - loss: 0.5719 - val_accuracy: 0.7300 - val_auc: 0.6485 - val_loss: 0.5754
Epoch 4/30
815/815 ━━━━━━━━━━━━━━━━━━━━ 6s 8ms/step - accuracy: 0.7302 - auc: 0.6513 - loss: 0.5725 - val_accuracy: 0.7304 - val_auc: 0.6476 - val_loss: 0.5711
Epoch 5/30
815/815 ━━━━━━━━━━━━━━━━━━━━ 7s 8ms/step - accuracy: 0.7298 - auc: 0.6556 - loss: 0.5713 - val_accuracy: 0.7303 - val_auc: 0.6502 - val_loss: 0.5705
Epoch 6/30
815/815 ━━━━━━━━━━━━━━━━━━━━ 7s 9ms/step - accuracy: 0.7308 - auc: 0.6578 - loss: 0.5709 - val_accuracy: 0.7309 - val_auc: 0.6513 - val_loss: 0.5714
Epoch 7/30
815/815 ━━━━━━━━━━━━━━━━━━━━ 6s 7ms/step - accuracy: 0.7299 - auc: 0.6600 - loss: 0.5703 - val_accuracy: 0.7298 - val_auc: 0.6529 - val_loss: 0.5708
Epoch 8/30
815/815 ━━━━━━━━━━━━━━━━━━━━ 6s 7ms/step - accuracy: 0.7313 - auc: 0.6600 - loss: 0.5699 - val_accuracy: 0.7303 - val_auc: 0.6505 - val_loss: 0.5722
Epoch 9/30
815/815 ━━━━━━━━━━━━━━━━━━━━ 7s 9ms/step - accuracy: 0.7304 - auc: 0.6631 - loss: 0.5693 - val_accuracy: 0.7295 - val_auc: 0.6534 - val_loss: 0.5704
Epoch 10/30
815/815 ━━━━━━━━━━━━━━━━━━━━ 8s 9ms/step - accuracy: 0.7313 - auc: 0.6571 - loss: 0.5700 - val_accuracy: 0.7295 - val_auc: 0.6521 - val_loss: 0.5715
```

```
Epoch 11/30
815/815 ━━━━━━━━━━━━━━━━━━━━ 6s 8ms/step - accuracy: 0.7323 - auc: 0.6623 - loss:
0.5692 - val_accuracy: 0.7287 - val_auc: 0.6566 - val_loss: 0.5708
Epoch 12/30
815/815 ━━━━━━━━━━━━━━━━━━━━ 6s 7ms/step - accuracy: 0.7310 - auc: 0.6643 - loss:
0.5686 - val_accuracy: 0.7294 - val_auc: 0.6513 - val_loss: 0.5721
Epoch 13/30
815/815 ━━━━━━━━━━━━━━━━━━━━ 6s 8ms/step - accuracy: 0.7317 - auc: 0.6659 - loss:
0.5682 - val_accuracy: 0.7312 - val_auc: 0.6548 - val_loss: 0.5707
Epoch 14/30
815/815 ━━━━━━━━━━━━━━━━━━━━ 7s 9ms/step - accuracy: 0.7308 - auc: 0.6665 - loss:
0.5686 - val_accuracy: 0.7293 - val_auc: 0.6545 - val_loss: 0.5715
Epoch 15/30
815/815 ━━━━━━━━━━━━━━━━━━━━ 6s 8ms/step - accuracy: 0.7316 - auc: 0.6656 - loss:
0.5684 - val_accuracy: 0.7300 - val_auc: 0.6553 - val_loss: 0.5703
Epoch 16/30
815/815 ━━━━━━━━━━━━━━━━━━━━ 6s 7ms/step - accuracy: 0.7317 - auc: 0.6661 - loss:
0.5684 - val_accuracy: 0.7312 - val_auc: 0.6561 - val_loss: 0.5713
Test accuracy: 0.7287  |  Test loss: 0.5708  |  Test AUC: 0.6566
Saved → dns_tunnel_transformer.keras

── Calibrating Ensemble Threshold ────────────────────────────
   Calibrated threshold: 0.3596  (F1=0.9630)
Saved → dns_threshold.json

── Final Ensemble Evaluation ─────────────────────────────────
              precision    recall  f1-score   support

      benign       0.98      0.98      0.98      7688
      attack       0.96      0.96      0.96      3485

    accuracy                           0.98     11173
   macro avg       0.97      0.97      0.97     11173
weighted avg       0.98      0.98      0.98     11173

Ensemble ROC-AUC: 0.9945

✅ Training complete. Artifacts saved:
   dns_tunnel_rf.pkl  |  dns_tunnel_transformer.keras  |  dns_scaler.pkl  |
dns_threshold.json
  ✅ Path A trained and saved to model_memory/

══ STEP 3 — Training Path B: Trafficformer (sequence reconstruction)
  ℹ  Learns WHAT normal traffic looks like (unsupervised)
```

```
  ℹ  Window size: 20 timesteps

[*] Loaded 37242 data points from /home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/data/merged_features_all.json
[*] Normalization params saved → model/dns_trafficformer_norm.json
[*] Train windows: 29778  |  Val windows: 7445
Model: "trafficformer_reconstruction"
 Total params: 4,270 (16.68 KB)
 Trainable params: 4,270 (16.68 KB)
 Non-trainable params: 0 (0.00 B)
Epoch 1/200
931/931 ━━━━━━━━━━━━━━━━━━━━ 20s 17ms/step - loss: 0.0085 - val_loss: 9.2889e-04
Epoch 2/200
931/931 ━━━━━━━━━━━━━━━━━━━━ 11s 12ms/step - loss: 0.0019 - val_loss: 3.0747e-04
Epoch 3/200
931/931 ━━━━━━━━━━━━━━━━━━━━ 12s 13ms/step - loss: 0.0011 - val_loss: 1.0832e-04
Epoch 4/200
931/931 ━━━━━━━━━━━━━━━━━━━━ 11s 12ms/step - loss: 7.8925e-04 - val_loss: 4.5351e-
05
Epoch 5/200
931/931 ━━━━━━━━━━━━━━━━━━━━ 13s 14ms/step - loss: 5.9682e-04 - val_loss: 3.8662e-
05
Epoch 6/200
931/931 ━━━━━━━━━━━━━━━━━━━━ 12s 12ms/step - loss: 4.8035e-04 - val_loss: 1.9175e-
05
Epoch 7/200
931/931 ━━━━━━━━━━━━━━━━━━━━ 22s 14ms/step - loss: 4.2162e-04 - val_loss: 1.8889e-
05
Epoch 8/200
931/931 ━━━━━━━━━━━━━━━━━━━━ 12s 13ms/step - loss: 4.0556e-04 - val_loss: 1.2425e-
05
Epoch 9/200
931/931 ━━━━━━━━━━━━━━━━━━━━ 14s 15ms/step - loss: 4.0140e-04 - val_loss: 1.2596e-
05
Epoch 10/200
931/931 ━━━━━━━━━━━━━━━━━━━━ 17s 18ms/step - loss: 3.9944e-04 - val_loss: 1.4044e-
05
Epoch 11/200
931/931 ━━━━━━━━━━━━━━━━━━━━ 15s 16ms/step - loss: 3.9414e-04 - val_loss: 1.5949e-
05
Epoch 12/200
931/931 ━━━━━━━━━━━━━━━━━━━━ 18s 19ms/step - loss: 3.9304e-04 - val_loss: 1.3471e-
05
Epoch 13/200
```

```
931/931 ━━━━━━━━━━━━━━━━━━━━ 14s 15ms/step - loss: 3.8981e-04 - val_loss: 1.0794e-
05
Epoch 14/200
931/931 ━━━━━━━━━━━━━━━━━━━━ 14s 15ms/step - loss: 3.9031e-04 - val_loss: 1.5223e-
05
Epoch 15/200
931/931 ━━━━━━━━━━━━━━━━━━━━ 15s 16ms/step - loss: 3.9409e-04 - val_loss: 2.0347e-
05
Epoch 16/200
931/931 ━━━━━━━━━━━━━━━━━━━━ 18s 20ms/step - loss: 3.9264e-04 - val_loss: 1.4055e-
05
Epoch 17/200
931/931 ━━━━━━━━━━━━━━━━━━━━ 18s 19ms/step - loss: 3.9384e-04 - val_loss: 1.2485e-
05
Epoch 18/200
931/931 ━━━━━━━━━━━━━━━━━━━━ 19s 21ms/step - loss: 3.9262e-04 - val_loss: 1.1960e-
05
Epoch 19/200
931/931 ━━━━━━━━━━━━━━━━━━━━ 18s 20ms/step - loss: 3.9467e-04 - val_loss: 1.2322e-
05
Epoch 20/200
931/931 ━━━━━━━━━━━━━━━━━━━━ 19s 20ms/step - loss: 3.9132e-04 - val_loss: 1.1362e-
05
Epoch 21/200
931/931 ━━━━━━━━━━━━━━━━━━━━ 20s 21ms/step - loss: 3.9445e-04 - val_loss: 9.5792e-
06
Epoch 22/200
931/931 ━━━━━━━━━━━━━━━━━━━━ 18s 19ms/step - loss: 3.9310e-04 - val_loss: 9.2342e-
06
Epoch 23/200
931/931 ━━━━━━━━━━━━━━━━━━━━ 17s 19ms/step - loss: 3.9349e-04 - val_loss: 1.1970e-
05
Epoch 24/200
931/931 ━━━━━━━━━━━━━━━━━━━━ 18s 19ms/step - loss: 3.9256e-04 - val_loss: 1.5175e-
05
Epoch 25/200
931/931 ━━━━━━━━━━━━━━━━━━━━ 17s 19ms/step - loss: 3.9287e-04 - val_loss: 1.1899e-
05
Epoch 26/200
931/931 ━━━━━━━━━━━━━━━━━━━━ 19s 20ms/step - loss: 3.9162e-04 - val_loss: 1.4101e-
05
Epoch 27/200
931/931 ━━━━━━━━━━━━━━━━━━━━ 21s 23ms/step - loss: 3.9337e-04 - val_loss: 1.4266e-
05
```

```
Epoch 28/200
931/931 ━━━━━━━━━━━━━━━━━━━━ 39s 21ms/step - loss: 3.9327e-04 - val_loss: 1.0512e-
05
Epoch 29/200
931/931 ━━━━━━━━━━━━━━━━━━━━ 18s 19ms/step - loss: 3.8884e-04 - val_loss: 1.3615e-
05
Epoch 30/200
931/931 ━━━━━━━━━━━━━━━━━━━━ 18s 20ms/step - loss: 3.9401e-04 - val_loss: 8.2018e-
06
Epoch 31/200
931/931 ━━━━━━━━━━━━━━━━━━━━ 17s 19ms/step - loss: 3.9360e-04 - val_loss: 1.3157e-
05
Epoch 32/200
931/931 ━━━━━━━━━━━━━━━━━━━━ 17s 18ms/step - loss: 3.9309e-04 - val_loss: 1.0399e-
05
Epoch 33/200
931/931 ━━━━━━━━━━━━━━━━━━━━ 18s 20ms/step - loss: 3.9307e-04 - val_loss: 1.2196e-
05
Epoch 34/200
931/931 ━━━━━━━━━━━━━━━━━━━━ 18s 19ms/step - loss: 3.9528e-04 - val_loss: 1.1198e-
05
Epoch 35/200
931/931 ━━━━━━━━━━━━━━━━━━━━ 18s 19ms/step - loss: 3.9090e-04 - val_loss: 1.3310e-
05
Epoch 36/200
931/931 ━━━━━━━━━━━━━━━━━━━━ 17s 19ms/step - loss: 3.9374e-04 - val_loss: 1.5163e-
05
Epoch 37/200
931/931 ━━━━━━━━━━━━━━━━━━━━ 18s 19ms/step - loss: 3.9321e-04 - val_loss: 1.6949e-
05
Epoch 38/200
931/931 ━━━━━━━━━━━━━━━━━━━━ 17s 19ms/step - loss: 3.9551e-04 - val_loss: 1.1813e-
05
Epoch 39/200
931/931 ━━━━━━━━━━━━━━━━━━━━ 21s 19ms/step - loss: 3.9090e-04 - val_loss: 1.4231e-
05
Epoch 40/200
931/931 ━━━━━━━━━━━━━━━━━━━━ 19s 20ms/step - loss: 3.9169e-04 - val_loss: 1.3664e-
05

[+] Model saved → model/dns_trafficformer.h5

── Calibrating Anomaly Threshold ─────────────────────────────
   Error stats → mean=0.0000  std=0.0000
```

```
   Anomaly threshold: 0.0000  (mean + 2.5×std)
[+] Threshold saved → model/dns_trafficformer_threshold.json

✅ Training complete.
   This model detects anomalies via RECONSTRUCTION ERROR.
   High error = unusual traffic pattern = potential DNS tunnel/attack.

   Next step: python Predict.py --filename <input> --save_name dns_trafficformer
  ✅ Path B trained and saved to model_memory/

══ Saving rich training memory
  Metadata   → /home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/model_memory/training_metadata.json
  Provenance → /home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/model_memory/training_provenance.json
  ✅ Training memory saved

══ Generating pipeline awareness check script
  ✅ Pipeline awareness check → /home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/model_memory/pipeline_awareness_check.sh
  ⚠ Run_Pipeline.sh not found at /home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/scripts/../Run_Pipeline.sh — patch skipped
  ℹ Manually add to your Run_Pipeline.sh after #!/bin/bash:
  ℹ    source "/home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/model_memory/pipeline_awareness_check.sh"

══════════════════════════════════════════════════════════════
  ✅  ML MODEL TRAINING COMPLETE  (13m 50s)
══════════════════════════════════════════════════════════════

Sources ingested:
  LOG files pre-processed  : 2
  PCAP files pre-processed : 0
  JSON features loaded     : 1
  CSV features loaded      : 0

Trained on:
  Total samples   : 37242
  Attack samples  : 11616
  Benign samples  : 25626
  Attack types    : unknown

Model memory saved to:  /home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/model_memory
```

```
  Random Forest          : dns_tunnel_rf.pkl
  Transformer            : dns_tunnel_transformer.keras
  Scaler                 : dns_scaler.pkl
  Threshold              : dns_threshold.json
  Trafficformer          : dns_trafficformer.h5
  Trafficformer thresh   : dns_trafficformer_threshold.json
  Provenance             : training_provenance.json
  Metadata               : training_metadata.json
  Training log           : /home/kali/Downloads/DNS-_raffic-
_Anomaly_Detection/logs/training_20260314_061518.log

  ▶  Run detection (will show training context):
  ./Run_Pipeline.sh

  Retrain on new data:
  ./Train_Models.sh --data-dir /new/captures --retrain
```